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Device, system and method for sublingual delivery

Abstract

A sublingual delivery device, system and method of operating such are provided. The sublingual delivery device comprises a device body comprising a delivery button configured to operate the sublingual delivery device, an ampoule carrying liquid, a threaded rod configured to push the ampoule towards a proximal end of the device body, a joints mechanism configured to convert vertical motion of the delivery button to horizontal motion of the threaded rod, an adjustment knob configured to set the volume of liquid to be delivered by adjusting a rotation degree of the knob, the knob located at a distal end of the device body, and a mouthpiece configured to be placed into a mouth of a user, the mouthpiece located at the proximal end of the device body, and the mouthpiece comprising a hollow needle configured to penetrate the ampoule and deliver liquid from the ampoule into the mouth of the user.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to sublingual delivery, and more specifically to a device, system and method for sublingual delivery of pharmaceutical compositions.

BACKGROUND

(2) Medications are used as treatment for many illnesses. The routes of medication administration, whether self-administered or delivered by a health care professional, may vary. Inappropriate administration may be dangerous, whereas, knowledge and understanding may improve safety and effectiveness of drug therapy. Most importantly, careful attention should be given to the dosage of any taken drug.

(3) The route of administration depends on three main factors: (i) the part of the body being treated; (ii) the way the drug is metabolized; and (iii) the formula of the drug. For instance, some drugs are degraded by stomach acid if they are taken through the mouth. Thus, such drugs may have to be administered by injection instead.

(4) In sublingual delivery, the drug, which may come in the form of tablets, films or liquid, is delivered along the floor of the mouth through the mucosal membrane linings. The sublingual route takes advantage of the permeability of the oral epithelium via passive diffusion into the lipoidal membrane. The absorption of a drug administered sublingually is 3-10 times greater than when administered through the oral route. In addition, because sublingual venous drainage is systemic rather than portal, hepatic first-pass elimination can be avoided (e.g., bypassing gastric acid and intestine). The rapid absorption and onset of action of the sublingual route compared to the oral route are particularly useful for treating breakthrough pain as well as other medical situations that require immediate onset of therapeutic effect, thus making the sublingual route a promising alternative with high efficacy. Moreover, systemic drug delivery through the sublingual route offers different populations of patients such as pediatric, geriatric, psychiatric or patients who suffer from dysphagia (difficulty to swallow) an accessible route for enhanced management and compliance with their treatment regimen.

(5) As explained hereinabove, sublingual administration of medicinal liquids is considered to be a promising alternative to the oral route. It is highly useful for rapid effect and is considered to be more patient compliant. However, current delivery systems and methods are imprecise and/or inconsistent and are mostly dependent on the administrator's technique, who might suffer from lack of dexterity or other physiological and physical limitations. Accidental dosing errors or inconsistent dosage are safety concerns that limit patient compliance with treatments and may result with poor health outcomes and even worsening of symptoms.

(6) An additional problem with current systems and methods lies with lack of efficient communication between patients and their physicians or other health practitioners. Typically, patients are given a treatment regimen to be followed for a certain period of time by their physician, which should be followed up with future appointments every predetermined time interval. Unreliable communication may compromise the ability of physicians to evaluate treatment effectiveness, which translates to poor patient's management and health outcomes. Moreover, for patients with chronic, progressive disease, ongoing adherence to prescribed treatment regimens is often challenging. Medication non-adherence may result with poor health outcomes or even substantial worsening of the patient's condition.

(7) It should also be noted that evidence is limited regarding the short- and long-term health effects of various liquid supplements and medicinal remedies, e.g., CBD/THC. Though the potential therapeutic benefits and use of CBD/THC is growing, there have not been enough largescale clinical trials that show, for example, that the benefits of the marijuana plant (as opposed to its cannabinoid ingredients) outweighs risks of using it in patients it is meant to treat. Data collection is important per a vast list of liquid medicines or supplements.

(8) Accordingly, user-friendly and accurate sub-lingual delivery systems for delivering liquid medicine are critically required as well as adjunctive solutions to enhance patient management and improve care. Specifically, there is a need for a hand-held device, designed to dispense precise dosing with the flexibility to make self-adjustments by the patient on the basis of treatment results, together with recordkeeping of timing, dosing and other data like self-reported symptoms, which will potentially enhance patient's compliance and health outcomes as well as improve care by

healthcare providers.

SUMMARY

(9) According to an aspect of some embodiments of the present disclosure, there is provided a system and method for easy and user-friendly sublingual administration, which may be operated by the patient, a care-giver, a physician or a healthcare provider. In some embodiments, the sublingual delivery system of the present disclosure may be used for repeated sublingual delivery (preferably self-delivery) of specific dosage of cannabis oil, though any other medication or therapeutic agent in liquid form may be delivered via the sublingual delivery system of the present disclosure.

(10) Sublingual products have been developed for numerous indications ranging from migraines, for which rapid onset of action is important, to mental illness, for which patient compliance is important for treating chronic indications such as depression and schizophrenia.

(11) Pharmaceutical preparations for sublingual administration are manufactured in various forms, e.g., (i) sublingual tablets, which are tablets that rapidly dissolve in the mouth of the patient with little or no residue, for example, Nitroglycerine tablets or anti-emetic ondansetron; (ii) sublingual strips, which are similar to tablets in that they easily melt in the mouth and dissolve rapidly, for example: Suboxone; (iii) multi-purpose tablets, which are soluble tablets for either oral, buccal or sublingual administration, also suitable for preparation of injections, for example, Hydrostat (hydromorphone) and a number of brands of morphine tablets and cubes; (iv) sublingual drops, which comprise a concentrated solution to be dropped under the tongue, for example, Nicocodeine cough preparations, THC, CBD, allergen immunotherapy, and vitamins; (v) sublingual spray, which is a spray to be applied under the tongue, for example, Fentanyl; and (vi) lozenge, which effects a metered and patient-controlled-rate combination of sublingual, buccal, and oral administration, for example, Actiq fentanyl lozenge-on-a-stick (e.g., a lollipop).

(12) Recent studies suggest that CBD may provide a safe, powerful natural treatment for many health issues. Sublingual is a popular choice for patients who would rather not inhale cannabis. For those who rule out vaping and smoking, either for personal health reasons or because they don't enjoy that type of administration route, sublingual administration offers the ideal substitute. It is one of the only non-inhaled options that can offer quick and long lasting (approximately 2-3 hours) relief from symptoms. Moreover, while edible cannabis effects might last throughout an entire day, the sublingual's effects will subside relatively quickly, allowing the patient to function and decide whether another dose is needed later along the day.

(13) The present disclosure describes a reusable device designed to repeatedly deliver adjustable, accurate dose of liquid medicine sublingually, with a support system for patient engagement, data collection, and analysis. The device may record doses and timing in order to generate a log of use, allowing patients to monitor their dosage and reliably report to their healthcare provider. Moreover, healthcare providers may be able to monitor patient's compliance via the communicated log. The delivery device may be uniquely designed for accurate dose self-sublingual delivery of liquid medicine, which may be loaded in an ampoule, with minimal waste. The sublingual delivery device of the present disclosure may be ergonomically designed to allow use by people suffering from, for example, low fine motor dexterity or be used by healthcare providers. The sublingual delivery system of the present disclosure may comprise a mechanical (manual), electromechanical, or digital dial for adjusting the required dosage. The sublingual system may further comprise a timer (e.g., to count up to 60 seconds or other time periods) to indicate patients of the time required for the medicine in the oral mucosa to facilitate absorption. In some embodiments, the sublingual device may further comprise a replaceable mouthpiece made from a biocompatible material, e.g., medical silicone comprising an inner hollow needle, which may comprise a thin hollow tube with one sharp tip. The hollow needle may be made of glass or any type of crystalline material. The hollow needle may be covered by the softer biocompatible mouthpiece to avoid gum damage and allow for proper flow of products with different viscosity and flow properties from the device into the sublingual area of a patient.

(14) In some embodiments, both the volume of dose delivered, as well as the timer may appear on a display, for example, a screen integrated on the device, e.g., an LCD screen. Volume accuracy of dosage may be measured by Linear variable differential transformer (LVDT) or linear and radial encoders with an accuracy of approximately 0.05 mm (which equals to 0.005 milliliter) or by a fine encoder that may be attached directly on the motor, in case of an electromechanical design, or on the main threaded rod, in case of a manual device. Another alternative may be to enable predetermined adjustable quantities that will initiate volume calculations to be presented on the screen. These features may also be controlled using a dedicated application on a computerized device, e.g., a mobile phone, a computer, a tablet, etc.

(15) In some embodiments, the ampoules holding the medicine or therapeutic agent may also be made of glass, or a type of crystalline material, to maintain the physiochemical properties of the medicament or therapeutic agent. Specifically, the hollow needle may penetrate the ampoule to deliver the medicine into the oral cavity. Reloading of the ampoule may be easy and straight forward.

(16) In some embodiments, the sublingual delivery device may be safely stored in an enclosed chargeable case, which may acquire, and store data collected by the device, monitor the temperature within the case in order to ensure proper storage temperature of the medicine carried within the sublingual delivery device. The case may even issue an alarm in case the temperature within it exceeds the maximum allowed temperature per medicine carried within the delivery device (i.e., the case and/or the device may provide temperature management).

(17) In some embodiments, the sublingual delivery device may comprise communication capabilities to report all collected data to an application software that may store the data. The sublingual delivery device's processor may have the capability to monitor consumption, timing, remaining volume in the ampoule and other characteristics of the medicament or therapeutic agent (e.g., its temperature, expiration date) and so on. The processor may also lock the device to avoid leakage of medication when not in use. In some embodiments, the processor may further control setting dosage of the medication. e.g., via the display of the sublingual device, or via the computerized device that is in communication with the sublingual device. Analyzing modules may provide information on trend and statistics of usage per user.

(18) In some embodiments, patients may manually record the status of their symptoms, via the dedicated application, in addition to data recorded automatically by the processor. All of the collected data may be stored in a cloud-based database and may be shared with healthcare providers through the Electronic Health Record (EHR) or research institutes. The collected data may be used to create largescale experimental groups to investigate treatment effectivity, improve personalized regimen, prediction algorithms, etc.

(19) One aspect of the present disclosure is a sublingual delivery device comprising a device body having a proximal end and a distal end, the device body comprising: a delivery button configured to operate the sublingual delivery device; an ampoule carrying liquid; a threaded rod configured to push the ampoule towards the proximal end of the device body; a joint mechanism configured to convert vertical motion of the delivery button to horizontal motion of the threaded rod; an adjustment knob configured to set the volume of liquid to be delivered by the sublingual delivery device by adjusting a rotation degree of the knob, the adjustment knob located at the distal end of the device body; and a mouthpiece configured to be placed into a mouth of a user, the mouthpiece located at the proximal end of the device body, and the mouthpiece comprising a hollow needle configured to penetrate the ampoule and deliver liquid from the ampoule into the mouth of the user.

(20) Optionally, the joint mechanism comprises a right vertical arm connected to a corresponding right arm, and a left vertical arm connected to a corresponding left arm. In some embodiments, the right and left arms are in turn connected to a ring arm that is horizontally pushed towards the distal end of the device body once the right and left vertical arms are vertically pushed by the delivery button.

- (21) Optionally, the device body further comprises a spring located between the adjustment knob and the threaded rod.
- (22) Optionally, the device body further comprises a display.
- (23) Optionally, the display displays the set volume of liquid to be administered by the sublingual delivery device.
- (24) Optionally, the display further displays a timer counting down a recommended time for holding the liquid in a sublingual cavity of the user, the recommended time is determined based on the type of liquid.
- (25) Optionally, the display displays the name of the liquid carried within the ampoule, the volume of liquid to be administered, the liquid volume carried by the ampoule, the volume remained in the ampoule, power source status of the sublingual delivery device, a timer, and/or any combination thereof.
- (26) Optionally, the device body further comprises a linear encoder configured to measure the location of the threaded rod, thereby to measure the amount of liquid that is delivered and the remaining volume of liquid carried within the ampoule.
- (27) Optionally, the device body further comprises a radial encoder configured to measure the rotation of the adjustment knob, thereby to measure the amount of liquid that is set to be delivered.
- (28) Optionally, the sublingual device is configured to provide an accurate volume of 0.05 mL with a variance of no more than 0.01 ml.
- (29) Optionally, the sublingual delivery device further comprises a processor configured to collect data related to time and volume of delivered liquid.
- (30) Optionally, the sublingual delivery device is configured to send the collected data to a computerized device in communication with said sublingual delivery device.
- (31) Optionally, the sublingual delivery device is controlled by a computerized device.
- (32) Optionally, the hollow needle is made of glass.
- (33) Optionally, the hollow needle is made of a crystalline material.
- (34) Optionally, the liquid carried within the ampoule may comprises one or more pharmaceutical compositions.
- (35) Another aspect of the present disclosure is a system for monitoring sublingual delivery, said system comprising: a sublingual delivery device as detailed hereinabove, a computerized device in communication with said sublingual delivery device; and a cloud-based database in communication with the computerized device, wherein the sublingual delivery device is configured to collect data, send it to the computerized device, and the computerized device is configured to send the data to be stored in the cloud-based database.
- (36) Optionally, the data comprises volume intake of the liquid, time of intake, remaining volume in ampoule, temperature or ampoule, expiration date of ampoule, user symptoms, user's physical state and/or any combination thereof.
- (37) Optionally, a physician or health care receives the collected data via the cloud-based database to monitor patient condition.
- (38) Another aspect of the present disclosure is a method for sublingual delivery, the method comprising: providing a sublingual delivery device as detailed above; setting a volume of liquid to be delivered by rotating the adjustment knob; placing the sublingual delivery device inside a sublingual cavity of the user; pressing the delivery button, thereby ejecting liquid from the mouthpiece through the hollow needle into the sublingual cavity; and receiving time and volume of delivered liquid via a computerized device in communication with the sublingual delivery device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Some non-limiting exemplary embodiments or features of the disclosed subject matter are illustrated in the following drawings.
- (2) In the drawings:
- (3) FIG. 1 is a schematic illustration of a sublingual delivery system, according to embodiments of the disclosure;
- (4) FIG. 2 is a schematic illustration of a sublingual delivery device, according to embodiments of the disclosure;
- (5) FIG. 3 is a schematic illustration of a sublingual delivery device without the upper cover, according to embodiments of the disclosure;
- (6) FIG. 4 is a schematic illustration of an exploded view of a sublingual delivery device, according to embodiments of the disclosure;
- (7) FIG. 5 is a schematic illustration of an upper view, side view and upper cross-section view of a sublingual delivery device, according to embodiments of the disclosure;
- (8) FIG. 6 a schematic illustration of a cross-section side-view of a sublingual delivery device, according to embodiments of the disclosure; and
- (9) FIG. 7 is a schematic flowchart illustrating a method for sublingual delivery, according to some embodiments of the present disclosure.
- (10) With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of embodiments of the disclosure. In this regard, the description taken with the drawings makes apparent to those skilled in the art how embodiments of the disclosure may be practiced.
- (11) Identical or duplicate or equivalent or similar structures, elements, or parts that appear in one or more drawings are generally labeled with the same reference numeral, optionally with an additional letter or letters to distinguish between similar entities or variants of entities, and may not be repeatedly labeled and/or described. References to previously presented elements are implied without necessarily further citing the drawing or description in which they appear.
- (12) Dimensions of components and features shown in the figures are chosen for convenience or clarity of presentation and are not necessarily shown to scale or true perspective. For convenience or clarity, some elements or structures are not shown or shown only partially and/or with different perspective or from different point of views.

DETAILED DESCRIPTION

- (13) Some embodiments of the present disclosure provide a user with the ability to control the volume or dose of a medicament or therapeutic agent to be taken sublingually. Currently, available delivery methods, such as syringes, droppers, or sprays, fall short because they either deliver imprecise or inconsistent doses, are difficult to self-administer, and/or do not track nor monitor the patient's adherence and clinical outcomes.
- (14) The present disclosure provides a sublingual delivery device, system and method for enabling delivery to the sublingual glands, that is easy to use, and which is configured to provide an accurate dose or volume of liquid, each time it is operated by a patient or any other user. Data regarding dosage intake, time of intake, and patient's symptoms and physical state may be collected via the sublingual delivery device and/or its supporting system, i.e., a computerized device communicating with the sublingual delivery device. The collected data may then be transferred to a cloud-based database in order to be shared with physicians or healthcare providers of the user.
- (15) Before explaining at least one embodiment of the disclosure in detail, it is to be understood that the disclosure is not necessarily limited in its application to the details of construction and the arrangement of the components and/or methods set forth in the following description and/or illustrated in the drawings and/or the Examples. The disclosure is capable of other embodiments or of being practiced or carried out in various ways.

(16) Reference is now made to FIG. 1, which is a schematic illustration of a sublingual delivery system, according to embodiments of the disclosure. System **100** for sublingual delivery may comprise sublingual delivery device **102**, which may be configured to provide accurate volume of liquid to be applied under the tongue of a user. The liquid may comprise one or more pharmaceutical compositions. Device **102** may comprise a mouth applicator to be inserted into the mouth of a user, e.g., user **104**. Device **102** may be connected to a software application via a computerized device **106**, which may be used to control and set various parameters on device **102** and collect info regarding the operation of device **102**. In some embodiments, the information collected by the application may be transferred via the computerized device **106** to a cloud-based database **108**, from which the information may be sent to the user's physician **110** upon user authorization, or may be sent to be part of a research **114**, for analyzing big data acquired from a large number of users of devices similar to device **102**, e.g., evaluating treatment's long-term efficiency. In some embodiments, the collected data per user may be processed into processed data **112**, which may be a log of usage of device **102** or may create individualized prediction models via machine learning and artificial intelligence algorithms, before it is sent to the physician **110** of the user **104** of device **102**.

(17) In some embodiments, device **102** may record the dosage (e.g., volume of liquid and timing) into the patient's log **112**. The Log **112** may be available locally on the device **102** and may be downloaded into other storages like a computerized device **106**, e.g., a personal computer or a mobile device, e.g., mobile phone, with or without designated software to read the data.

(18) In some embodiments, log **112** upon user approval and registration, may be available on the patient's EHR (Electronic Health Record) to allow accessibility for the healthcare provider **110** in order to track the patient or user's progress. The data the healthcare provider **110** may be able to receive may comprise: a complete record of all taken dosages, alarms on anomalies in the general trend, correlation between the patient's data and a global data-base use case with the same case parameters.

(19) Reference is now made to FIG. 2, which is a schematic illustration of a sublingual delivery device, according to embodiments of the disclosure. In some embodiments, sublingual delivery device **102** may be ergonomically designed, while enabling self-handling and self-operation of users, even users who are challenged with fine motoric skills and/or suffer from functional disabilities, such as tremor. In some embodiments, the body **201** of device **102** may be made from stainless steel and/or aluminum, giving it durability and an esthetic appearance. In addition, stainless steel and aluminum have the advantage of being sustainable materials that are environmentally friendly. However, in other embodiments, body **201** may be made of other materials.

(20) According to some embodiments, sublingual delivery device **102** may comprise a manual knob comprising dial **202** that may be manually turned by a user of sublingual delivery device **102** in order to adjust the proper volume or dosage to be taken by the user. In other embodiments, instead of a manual dial **202**, sublingual delivery device **102** may comprise a digitally operated and controlled dial or other digitally operated dosage determining element. In other embodiments, sublingual delivery device **102** may comprise a linear or radial actuator.

(21) The volume may be set up by the patient or user by turning the knob comprising a dial **202**, which shows the dosage to be applied by the user. In case of an electromechanically controlled mechanism, the user can set the volume via the application operating on computerized device **106**.

(22) The sublingual delivery device **102** may provide accurate volume, for example, 0.05 mL = 1/20 of Drop, with consistent repeatability and low variance, for example, variance of no more than 0.01 ml. Sublingual delivery device **102** may comprise a mouthpiece **204**, which may be of a shape that conforms to the shape of a user and may be made from a relatively soft biocompatible material, since it is inserted into the mouth of a user, e.g., medical grade silicon. The material from which mouthpiece **204** may be made of, should be configured to prevent bacteria growth, e.g., medical

grade silicon, and should be soft to protect the gum and sublingual surface of the patient.

(23) In some embodiments, sublingual delivery device **102** may comprise a delivery button **206** that is to be operated by a user. e.g., by pushing, moving or pressing button **206**. Once button **206** is operated, while mouthpiece **204** of sublingual delivery device **102** is placed into the mouth of a patient, the adjusted amount of liquid may be passed through the sublingual delivery device **102** and out through the mouthpiece **204** into the sublingual glands of a user. In some embodiments, device **102** may comprise a display **208**, which may be an LCD screen or LED matrix. Display **208** may display the adjusted dosage **218** or volume quantity to be applied, as well as the remaining volume within device **102**. Screen **208** may further display a timer **228** and any other required information. Immediately at the time of the delivery, the sublingual delivery device **102** may activate a timer **228**, e.g., Of 60 seconds, though other time periods may be implemented per type of liquid. The timer **228** may count the duration recommended to hold the medicine under the tongue for maximal absorbance and increased bioavailability.

(24) In some embodiments, device **102** may comprise a power source **230**. Power source **230** may be placed within device body **201** and may either be recharged by wires or wireless. In some embodiments, power source **230** may comprise batteries, which may or may not be rechargeable.

(25) In some embodiments, sublingual delivery device **102** may comprise communication capabilities to report all collected data to an application software that may store the data.

Sublingual delivery device **102** may comprise a processor **240**, which may have the capability to monitor consumption, timing, remaining volume in the ampoule and other characteristics of the medicament of therapeutic agent (e.g., its temperature, expiration date) and so on. The processor **240** may lock the device to avoid leakage of medication when not in use. In some embodiments, the processor **240** may further control setting dosage of the medication, e.g., via the display of the sublingual device, or via the computerized device that is in communication with the sublingual device. Analyzing modules may provide information on trend and statistics of usage per user.

(26) Reference is now made to FIG. **3**, which is a schematic illustration of a sublingual delivery device without the upper cover, according to embodiments of the disclosure. Sublingual delivery device **102** may comprise, in addition to the manual dial **202**, a plunger **302**, which may be connected to a joints mechanism **304**. Joints mechanism **304** may convert vertical movement of the user operating button **206** to horizontal movement of plunger **302**. Sublingual delivery device **102** may further comprise an ampule **306** carrying the liquid medicament or therapeutic agent, from which liquid may be pushed out at the desired amount predetermined by the user via dial **202**, through mouthpiece **204** and into the mouth cavity to be taken sublingually.

(27) According to some embodiments, sublingual delivery device **102** may be carried and protected by a case or carrier (not shown). The case configured to carry sublingual delivery device **102** may comprise a first section into which device **102** may be placed and a second section into which ampule **306** may be inserted. The shapes of the first and second sections may be configured such that they hug the respective elements positioned within, i.e., device **102** and ampule **306**. In some embodiments, the case or carrier of sublingual delivery device **102** may comprise charging capability, meaning that sublingual delivery device **102** may be charged while being positioned within its carrying case. In some embodiments, the carrying case may comprise temperature monitoring to monitor environment temperature such to ensure the ampule is held in proper low temperature that would not affect the chemical properties and efficacy of the liquid carried by ampule **306**. In some embodiments, the carrying case may further comprise an alert system to provide notifications to the user, for example, in case there is an issue with respect to measured environment temperature, or to indicate that sublingual delivery device **102** is fully charged or when it is out of power. In some embodiments, the carrying case may be connected to a communication network and may allow transfer of information stored on the sublingual delivery device **102** to a computerized device, e.g., mobile phone **106** (FIG. **1**), to the healthcare provider **110** (FIG. **1**) or any other external receiver.

(28) Reference is now made to FIGS. 4-6, which are schematic illustrations of an exploded view, an upper view, side view, upper cross-section view and a cross-section side-view of a sublingual delivery device, according to embodiments of the disclosure.

(29) In some embodiments, sublingual delivery device **400** may be similar to sublingual delivery device **102** and may comprise similar components, such as a processor and a power source, in addition to the components detailed hereinbelow. In some embodiments, sublingual delivery device **400** may comprise a body base part **501**, which may be connected to a body cover part **502** in order to create a closed body or housing, e.g., body **201** of device **102** (FIG. 2). Device **400** may comprise a dose adjustment knob **503** (similar to knob **202**) and a delivery pressing button **504** (similar to button **206**). Following adjustment of the required dose by adjustment knob **503**, the delivery button **504** may be pressed to deliver the adjusted dose. Sublingual delivery device **400** may comprise a display **505** (similar to display **208**), which may be, for example, an LCD screen. Display **505** may be powered by a power source, similar to power source **230** (FIG. 2). Display **505** may display various parameters on it, e.g., name of the liquid carried by the sublingual delivery device **400**, volume of liquid to be delivered by device **400**, volume remaining in device **400**, a timer to indicate recommended time during which the liquid should maintain in the sublingual cavity, and power source status, i.e., how much power device **400** still has, or whether the power source is high or low.

(30) Device **400** may comprise a piston and/or a threaded rod **507** which may be connected to an equilibrium spring **508** via a joints mechanism **404** (similar to joints mechanism **304**, FIG. 3), which comprises elements **509**, **515**, **516** and **517**. Joints mechanism **404** may convert vertical movement of delivery button **504**, for example, when delivery button **504** is pressed vertically downwards, to horizontal movement of threaded rod or piston **507**, which may comprise a plunger **514**, towards the proximal end of device **400**. Joints mechanism **404** may comprise ring arm **509** on one end and may further comprise left and right vertical arms **516** on an opposite end. Joints mechanism **404** may comprise a right side that is identical to its left side. Joints mechanisms **404** may comprise left arm **515**, which may connect one side, e.g. the left side, of ring arm **509** to a left vertical arm **516**. Left arm **515** may be connected to left vertical ring **516** via rotational joint **506** and may be connected to arm **509** via rotational joint **526**. Joints mechanism **404** may further comprise a right arm **517** (located behind left arm **515**) that may connect another side, e.g., the right side, of ring arm **509** to a right vertical arm **516** (that is located behind left vertical arm **516**). Right arm **517** may be connected to left vertical ring **516** via rotational joint **506** and may be connected to arm **509** via rotational joint **526**.

(31) Once delivery button **504** is vertically pressed, left and right vertical arms **516** are pushed down, thereby pulling corresponding left arm **515** and right arm **517** down and towards the proximal end of device **400**, which in turn pull ring arm **509** towards the proximal end of device **400**. Ring arm **509** then pulls piston and/or threaded rod **507** as well as adjustment knob **503** along with it towards the proximal end of device **400**.

(32) In some embodiments, left and right vertical arms **516** may convert the pressing vertical movement of delivery button **504** to horizontal movement of piston and/or threaded rod **507** towards the proximal end of device **400**, where an ampoule **510** which carries liquid is located. The liquid may comprise one or more pharmaceutical compositions. The liquid carried within ampoule **510** may be configured to be sublingually delivered to a user who is operating device **400**, through a mouthpiece **511**. Thus, operating delivery button **504** results in delivery of the liquid carried by ampoule **510**.

(33) In some embodiments, device **400** may comprise a mouthpiece **511** (similar to mouthpiece **204**, FIG. 2), which may be made of a biocompatible material, e.g., medical grade silicon. Mouthpiece **511** may comprise a hollow needle **512**, e.g., a glass hollow needle, though other materials, e.g., crystalline materials, may be used to manufacture hollow needle **512**. In some embodiments, when ampoule **510** is inserted into sublingual delivery device **400**, hollow needle

512 may be configured to puncture a hole through the proximal end of ampoule **510**. e.g., through a flexible ampoule cap **550**, to allow exit of liquid therethrough.

(34) In yet other embodiments, instead of mouthpiece **511** comprising hollow needle **512** configured to puncture an ampoule cap **550**, ampoule **510** may comprise a one-way valve (not shown) located at its proximal end, such that when moveable ampoule seal **513** is pushed by plunger **514** into ampoule **510**, the pressed liquid within ampoule **510** would cause the one-way valve to open and enable the predetermined dose of liquid to exit through it and thus to exit from device **400**.

(35) In some embodiments, after adjustment knob **503** is turned in the proper amount which is equivalent to delivering a predetermined dose of liquid, delivery button **504** may be vertically pushed or pressed and joint mechanism **404** may convert the vertical movement of delivery button **504** to horizontal movement of piston and/or threaded rod **507** towards the proximal end of device **400**, as explained hereinabove. When piston and/or threaded rod **507** is pushed towards the proximal end of device **400**, it may push plunger **514** towards ampoule **510**, thereby causing ampoule seal **513**, which is located at the distal end of ampoule **510** and also adjacent to plunger **514**, to be pushed into ampoule **510** in the exact amount dictated by adjustment knob **503**. When ampoule seal **513** is pushed into ampoule **510**, the predetermined dose of liquid is forced to exit through hollow needle **512** and into the sublingual cavity of a user.

(36) In some embodiments, hollow needle **512** may be used to penetrate the ampoule cap **550** and may allow low friction and smooth flow of various liquids, even viscous liquids like CBD or THC oil. In some embodiments, the length of hollow needle **512** may be approximately 10-15 mm, i.e., approximately between 0.4-0.6 inches, with an inner diameter of between 0.5 mm-1 mm. In some embodiments, the volume that may be trapped in the hollow needle **512** may be between 0.008-0.012 ml, though other dimensions and thus other volumes may be implemented.

(37) In some embodiments, the medicinal liquid, for example, CBD\THC oil may be stored in ampoule **510**. In some embodiments, ampoule **510** may be made of glass, though other materials may be used.

(38) In some embodiments, instead of via a needle, another way of connecting ampoule **510** to mouthpiece **511** is via a spring with a conical cup.

(39) According to some embodiments, there may be several options for refilling the sublingual delivery device **400** (or sublingual delivery device **102**). Among them, to replace a used and empty ampoule **510** with a new one, such that each ampoule is for single use or is intended for several doses, e.g., between 20 to 50 doses, after which ampoule **510** is discarded; or to refill the ampoule, using, for example, a syringe that contains the same prescribed liquid, thereby causing the ampoules to be reusable.

(40) In some embodiments, instead of a manual mechanical mechanism of operation of sublingual delivery device **400**, sublingual delivery device **400** may comprise an electromechanical mechanism to be initiated by pressing an operation button. A small motor or linear actuator may then press plunger **514**, thereby applying the liquid carried by ampoule **510**. In such embodiments, there may be no need for applied force in pressing the operation button.

(41) In some embodiments, the method of operating sublingual delivery device **400** may comprise several stages; (i) a loading stage, (ii) a dose setting stage, (iii) a delivering stage and (iv) a storage and data collection stage.

(42) According to some embodiments, device **400** may be loaded with a liquid carrying element, e.g., ampoule **510**. Device **400** may comprise several encoders, e.g., a linear encoder **521** that may measure the location of the threaded rod or piston **507** and thus measures the amount of liquid that is delivered and may further measure the current or remaining volume of liquid carried within the ampoule, and a radial encoder **520**, which may measure the rotation of the adjustment knob **503** and thus measures the amount of liquid that is set to be delivered.

(43) According to some embodiments, linear encoder **521** may measure the location of threaded rod

or piston **507** relative to the location of body **501** and of knob **503**. Accordingly, the location of threaded rod **507** with respect to ampule **510** may be determined. i.e., how much threaded rod **507** is pressed into ampule **510**, which may indicate the amount of fluid there is in ampule **510** since the initial amount carried by ampule **510** is known.

(44) In some embodiments, encoders implemented as part of device **400**, such as linear encoder **521** and radial encoder **520**, may be used for calibrating device **400** as well as for routine checks.

(45) In some embodiments, radial encoder **520** may measure the number of turns knob **503** is rotated by a user. Radial encoder **520** may measure the degrees at which knob **503** is turned and positioned at, at any time. The measured degrees may indicate the volume that may be delivered by calculating the length that threaded rod or piston **507** would undergo when turned by that measured degrees, depending on the pitch of the threads of threaded rod **507** and the diameter of the ampule **510**. For example, if the thread pitch is 1 mm, a 360 degrees rotation may be equivalent to an increment of 1 mm. 1 milliliter equals 1 cubic centimeter, which equals 1000 cubic millimeter. If, for example, the diameter (D) of the ampule, e.g., ampule **510**, is 12 mm, then the area of the ampule, which equals $2D/4 \times \pi$ is a total of 113 square millimeter. Thus, in this example, when rotating knob **503** at 360 degrees, threaded rod **507** may push or apply pressure against plunger **514** such to move 1 mm forward towards ampule **510**. A 1 mm advancement against ampule **510** may be, for example, equivalent to the volume of 0.113 ml, by using the simple volume calculating equation of: $\text{Volume} = H \times \text{Area}$, whereby H denotes the height, i.e., the distance ampule **510** is pushed at. Accordingly, the volume of liquid that is provided when ampule **510** is pushed by 1 mm (by turning knob **503** at 360 degrees) equals: $\text{volume} = H \times \text{Area} = 1 \times 113 \text{ square millimeter} = 0.113 \text{ milliliter}$. This is of course only one example, and different volumes may be provided when turning the knob **503** at 360 degrees or less, or for turning knob **503** more than one turn of 360 degrees, further depending on the diameter of the ampule, and the pitch of thread of the threaded rod **507**.

(46) In some embodiments, before placing an ampoule, e.g., ampoule **510** into device **400**, there may be an ampoule scanning step, e.g., an RF ID ampoule scanning step, a QR code scanning step or scanning any other type of code, to identify contents of the ampoule and to validate that the selected ampoule is the one to be inserted into device **400**, per a specific user. Scanning the ampoule tag may also provide information with respect to setting parameters of the sublingual delivery device **400** per the characteristics of the contents of the ampoule. For example, the type of medicinal liquid carried within the ampoule, the viscosity of the liquid, the temperature the ampoule should be stored at, the expiration date of the ampoule, and so on.

(47) During the loading stage of device **400**, the threaded rod **507** should be fully screwed into the dose adjustment knob **503**, which may or may not comprise a dial. In case the threaded rod **507** is not fully inserted in the adjustment knob **503**, the knob **503** should be turned clockwise until the threaded rod **507** is fully inserted into it and the knob **503** will reach maximum length, while pulling in the threaded rod **507**. At this point, display **505** may display a word indicating it is the beginning of the operation process of device **400**. For example, the word 'HOME' or 'STRAT' may appear on display **505**.

(48) During the volume or dose setting stage, the operating threaded rod **507** may be held by two keys or protrusions **530a** and **530b** located on both sides of threaded rod **507**, which may slide along slots (not shown) that may be located on the inner sides of device body **501** and device cover **502**. These keys **530a** and **530b** prevent threaded rod **507** from rotating. As knob **503** is turned, it may advance longitudinally in the direction away from ampule **510**, i.e., screwing itself out of the body **501** of device **102**, thereby pulling the joint mechanism **520**, which may lift the button **504** upwards. The amount of liquid to be delivered may be measured by radial encoder **520**, which may be an optical encoder. The amount to be delivered may be displayed on the screen **505**.

(49) At this initial state of determining dose, the user may load the ampule **510** into the device and attach the mouthpiece **511**, which comprises a hollow needle **512**, onto the device **400** next to the ampule **510**.

(50) The user may then turn the adjustment knob **503** counterclockwise and press the delivery button **504** until a small drop appears to come out of the mouthpiece **511**. The device **400** is now loaded and ready to be used.

(51) During the delivery stage, delivery button **504** is pressed, pushed or otherwise moved such to cause the joints mechanism **404** and adjustment knob **503** to be pulled forward towards the loaded ampule **510**, thereby pressing and pushing the threaded rod **507** against the plunger **514** which pushes the ampule seal **513** into ampoule **510**. Pushing ampoule seal **513** into ampoule **510** results in pushing the predetermined dose of liquid from the liquid carried by ampule **510** out of ampule **510** through the (glass) hollow needle **512**. When a user operates button **504**, e.g., by pressing it or pushing it, left and right vertical arms **516** are pushed down towards the device body **501**. Left vertical arm **516** may be connected to left arm **515** and right vertical arm **516** may be connected to right arm **517** via a corresponding rotational joint **506**. When vertical arms **516** are pushed down towards device body **501**, vertical arms **516** pull left arm **515** and right arm **517** along with it towards the proximal end of device **400**. Left arm **515** and right arm **517** may be further connected to ring arm **509** via a second rotational joint **526**. Once left arm **515** and right arm **517** are pulled towards the proximal end of device **400**, they pull ring arm **509** along with them. Ring arm **509** is pulled horizontally towards the proximal end of device **400**, i.e., towards mouthpiece **511**, piston and/or threaded rod **507** that is attached to ring arm **509** is caused to move with ring arm **509**, thereby pushing threaded rod **507** against plunger **514**, which leads to the release of the predetermined or selected dose of liquid out of ampule **510** through hollow needle **512** and into the sublingual cavity of the user, when mouthpiece **511** is located within such cavity.

(52) The device **400** may notify the user when complete dose or liquid volume has been delivered and a timer (e.g., timer **228**. FIG. 2) may be triggered for a predetermined time period depending on the type of liquid being administered. For example, the timer may count 60 seconds, though other time periods may be implemented.

(53) In some embodiments, the user may choose the dose to be delivered by turning the knob **503** counterclockwise or clockwise.

(54) In some embodiments, the user may place the mouthpiece **511** in the sublingual cavity and then operate device **400**, e.g., by pressing the delivery button **504**.

(55) In some embodiments, a notification sound may be generated once the volume was delivered in full, which may initiate the timer. Once notified, the user may hold the device downwards and may view screen **505**, displaying the countdown of the time period recommended for holding the liquid sublingually for proper absorption. For example, sixty seconds is often the recommended duration to hold the liquid in the sublingual cavity for it to be well absorbed.

(56) During storage and data collection stage, in order to maintain the integrity of the ampule contents, the user may store the device in a carrying case, with a temperature sensor that may measure environmental temperature and may indicate when temperature is too high, too low, or adequate. The carrying case may also serve as a docking station with communication capabilities that receives and stores data collected by the device **400**, and further transmits the data to a software application and stores it on a cloud database. In some embodiments, the carrying case may further be used to supply power to device **400** in case it includes electromechanical components.

(57) In some embodiments, device **400** may display trends and other statistics, along with a log of doses and timing per user following processing of information, e.g., via analysis modules. Users or patients may also record descriptive information regarding their symptoms or general feeling via a computerized device, e.g., computerized device **106** (FIG. 1), such as a mobile phone or tablet. This may allow patients to monitor their usage of the liquid medicament or therapeutic agent that was prescribed to them (i.e., enhance adherence to treatment) as well as report reliable information to their healthcare provider (i.e., improve patient management and care).

(58) Reference is now made to FIG. 7, which is a schematic flowchart illustrating a method for sublingual delivery via a sublingual delivery device, according to some embodiments of the present

disclosure. In some embodiments, a method **700** for sublingual delivery may comprise operation **702**, which may comprise removing a sublingual delivery device, e.g., device **102**, or device **400** from its carrying case, and turning it on, e.g., by pressing a power button. It is then to be determined in operation **704** whether the ampule carried within the device, e.g., ampule **510**, is empty. When the ampule is empty, a procedure for replacing the empty ampule with a new full ampule **706** takes place. When the ampule is not empty, method **700** may comprise operation **708**, which may comprise rotating the knob, e.g., knob **503**, counterclockwise, thereby setting the amount of medicament or therapeutic agent to be delivered. Method **700** may further comprise operation **710** comprising applying and aiming the mouthpiece, e.g., mouthpiece **511**, toward the sublingual cavity of a user.

(59) In some embodiments, method **700** may further comprise operation **712** comprising operating the sublingual delivery device, e.g., by pressing a button, e.g., button **504**, down or inwards towards device body **501**, thereby ejecting the liquid of the medicament or the therapeutic agent from the delivery device. According to some embodiments, method **700** may comprise operation **714**, which may comprise storing the time of intake and the volume of liquid that was delivered by the device, e.g., device **102** or device **400**.

(60) In some embodiments, a user may require an additional dose as in operation **716**. When a user requires an additional dose, method **700** continues by repeating operations **708** to **712** and operation **714**. When the user does not require an additional dose, method **700** may comprise operation **718** comprising storing the device back in the carrying case.

(61) In some embodiments, at the end of the delivery procedure, either after operation **714** or after operation **718**, method **700** may comprise operation **720** comprising transferring the data collected during operation **714** to an external receiver, e.g., the user's mobile app or any other medium that may enable sharing the collected information regarding the delivery session with the user's caregiver. Method **700** may further comprise operation **722** comprising turning the delivery device off.

(62) The flowchart and block diagrams illustrate architecture, functionality or an operation of possible implementations of systems, and methods according to various embodiments of the present disclosed subject matter. It should also be noted that, in some alternative implementations, illustrated or described operations may occur in a different order or in combination or as concurrent operations instead of sequential operations to achieve the same or equivalent effect.

(63) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprising”, “including” and/or “having” and other conjugations of these terms, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(64) The terminology used herein should not be understood as limiting, unless otherwise specified, and is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosed subject matter. While certain embodiments of the disclosed subject matter have been illustrated and described, it will be clear that the disclosure is not limited to the embodiments described herein. Numerous modifications, changes, variations, substitutions and equivalents are not precluded.

Claims

1. A sublingual delivery device comprising: a device body having a proximal end and a distal end, said device body comprising: a delivery button configured to operate the sublingual delivery device; an ampoule carrying liquid comprising an ampoule seal; a threaded rod configured to push the ampoule seal towards the proximal end of the device body; a joints mechanism configured to convert vertical motion of the delivery button to horizontal motion of the threaded rod; an adjustment knob configured to set the volume of liquid to be delivered by the sublingual delivery device by adjusting a rotation degree of the knob, said adjustment knob located at the distal end of said device body; and a mouthpiece made of a biocompatible material configured to be placed into and have a shape that conforms to the shape of a mouth of a user, said mouthpiece located at the proximal end of said device body, and said mouthpiece surrounding a hollow needle configured to penetrate the ampoule and deliver liquid from the ampoule into the mouth of the user; wherein said joints mechanism comprises a ring arm, left and right vertical arms, and left and right arms, said ring arm connected to the left and right arms via a first corresponding joint, and said left and right arms are connected to the left and right vertical arms via a second corresponding joint, wherein actuation of the delivery button is restricted to a linear movement in a vertical direction, and when the delivery button is pressed vertically downwards, the left and right vertical arms are pushed down, thereby pulling the left and right arms down and towards the proximal end of the device body, which in turn pulls the ring arm towards the proximal end of the device body, to thereby pull the threaded rod towards the proximal end of the device body, further wherein the entire joints mechanism is encapsulated within the device body.
2. The sublingual delivery device of claim 1, wherein said right vertical arm is connected to the corresponding right arm, and the left vertical arm is connected to the corresponding left arm, wherein said right and left arms are in turn connected to the ring arm that is horizontally pushed towards the distal end of the device body once the right and left vertical arms are vertically pushed by the delivery button.
3. The sublingual delivery device of claim 1, wherein said device body further comprises a spring located between the adjustment knob and the threaded rod.
4. The sublingual delivery device of claim 1, wherein said device body further comprises a display.
5. The sublingual delivery device of claim 4, wherein said display displays the set volume of liquid to be administered by the sublingual delivery device.
6. The sublingual delivery device of claim 5, wherein said display further displays a timer counting down a recommended time for holding the liquid in a sublingual cavity of the user, said recommended time is determined based on the type of liquid.
7. The sublingual delivery device of claim 4, wherein said display displays the name of the liquid carried within the ampoule, the volume of liquid to be administered, the liquid volume carried by the ampoule, the volume remained in the ampoule, power source status of the sublingual delivery device, a timer, and/or any combination thereof.
8. The sublingual delivery device of claim 1, wherein said sublingual device is configured to provide an accurate volume of 0.05 mL with a variance of no more than 0.01 ml.
9. The sublingual delivery device of claim 1, wherein said sublingual delivery device further comprising a processor configured to collect data related to time and volume of delivered liquid.
10. The sublingual delivery device of claim 9, wherein said sublingual delivery device is configured to send the collected data to a computerized device in communication with said sublingual delivery device.
11. The sublingual delivery device of claim 1, wherein said sublingual delivery device is controlled by a computerized device.
12. The sublingual delivery device of claim 1, wherein said hollow needle is made of glass.
13. The sublingual delivery device of claim 1, wherein said hollow needle is made of a crystalline material.

14. The sublingual delivery device of claim 1, wherein said liquid comprises one or more pharmaceutical compositions.

15. The sublingual device of claim 1, wherein said mouthpiece is soft to protect the gum and sublingual surface of the mouth of the user.

16. A system for monitoring sublingual delivery, said system comprising: a sublingual delivery device according to claim 1; a computerized device in communication with said sublingual delivery device; and a cloud-based database in communication with the computerized device, wherein said sublingual delivery device is configured to collect data, send it to the computerized device, and said computerized device is configured to send the data to be stored in the cloud-based database.

17. The system of claim 16, wherein said data comprises volume intake of the liquid, time of intake, remaining volume in ampoule, temperature of ampoule, expiration date of ampoule, user symptoms, user's physical state and/or any combination thereof.

18. The system of claim 16, wherein the collected data stored in the cloud-based database is configured to be sent to a physician or health care provider to monitor patient condition.

19. A method for sublingual delivery, said method comprising: providing a sublingual delivery device according to claim 1; setting a volume of liquid to be delivered by rotating the adjustment knob; placing the sublingual delivery device inside a sublingual cavity of the user; pressing the delivery button, thereby ejecting liquid from the mouthpiece through the hollow needle into the sublingual cavity; and receiving time and volume of delivered liquid via a computerized device in communication with the sublingual delivery device.

20. The sublingual delivery device of claim 1, wherein said device further comprises a linear encoder configured to measure the location of the threaded rod, thereby to measure the amount of liquid that is delivered and the remaining volume of liquid carried within the ampoule.

21. The sublingual delivery device of claim 1, wherein said device further comprises a radial encoder configured to measure the rotation of the adjustment knob, thereby to measure the amount of liquid that is set to be delivered.
